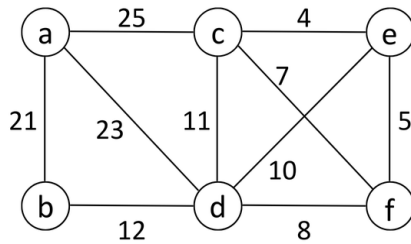


COM3067 ALGORITHMS

FINAL EXAM QUESTIONS – 12.01.2026

1. (18p) Consider the undirected, weighted graph shown below.



You are running Prim's algorithm to obtain the minimum spanning tree of the given graph. During the execution of the algorithm, what are the intermediate values of the key and the parent parameters for the node C and E.

2. (30p) Given a sequence of n houses $X = x_1, \dots, x_n$ where x_i represents the value of the i -th house. Design a dynamic programming that finds the maximum amount of money that can be robbed. The constraint is that two adjacent houses cannot be robbed simultaneously.

(a) (5p) Define a subproblem.

(b) (15p) Construct the recurrence relation.

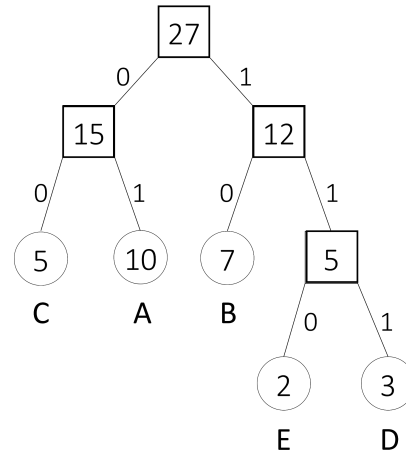
(c) (5p) Determine base cases.

(d) (5p) Fill in the dynamic programming table given below.

i	1	2	3	4	5
x_i	7	4	11	2	9
OPT(i)					

3. (20p) Given two sorted arrays of equal size n , find the median of the union of the two arrays in $O(\log n)$ time using a divide and conquer approach.

4. (20p) A prefix tree is provided for character frequencies of a string.



(a) (7p) Calculate the total cost of the encoding of the string considering the given tree.

(b) (13p) Construct a Huffman tree using the same string, then calculate the total cost and compare it with the prefix cost.

5. (12p) A depth-first search has been performed on a graph. The discovery and finishing times of its vertices are given as A(1, 14), B(3, 4), C(10, 13), D(2, 5), E(6, 9), F(7, 8), and G(11, 12). Using only this information, construct an adjacency list for a possible graph.